

Mini Job Board: Database Design Specification

This document provides a comprehensive overview of the database architecture for the Mini Job Board application. It details the relational schema, entity relationships, and specific table structures implemented with MySQL and managed via JPA annotations.

Entity Relationship Overview

The system is designed around a central User entity that interacts with various professional and recruitment-focused modules. The primary relationships are defined as follows:

- **User Centrality:** A user can have many experiences, educations, certifications, and skills.
- **Recruitment Flow:** Employers (Users with the EMPLOYER role) post job listings, while Candidates (Users with the CANDIDATE role) submit applications to those jobs.
- **Social & Engagement:** Users can receive multiple recommendations and save various job listings for future reference.
- **Job Management:** Each job listing can receive multiple candidate applications.

Detailed Table Schemas

1. Users Table

The users table acts as the primary repository for authentication and profile metadata for both recruiters and seekers.

Field	Type & Constraints	Description
id	BigInt, PK, Auto-inc ▾	Unique identifier.
email	VarChar(100), Uniq... ▾	Login and contact email.
password	VarChar(100) ▾	BCrypt hashed password.
full_name	VarChar(100) ▾	User's legal full name.
role	VarChar(20) ▾	CANDIDATE or EMPLOYER.
headline	VarChar(255) ▾	Professional tagline.
location	VarChar(255) ▾	Geographical location.
about	Text ▾	Professional biography.
avatar_url	VarChar(255) ▾	URL for profile photo.
banner_url	VarChar(255) ▾	URL for profile banner.

2. Jobs Table

This table stores job advertisements and their associated metadata.

Field	Type & Constraints	Description
id	BigInt, PK, Auto-inc ▾	Unique listing identifier.
employer_id	BigInt, FK (users) ▾	Posting recruiter reference.
title	VarChar(255) ▾	Job role title.

Field	Type & Constraints	Description
company	VarChar(255) ▾	Company name.
workplace_type	VarChar(20) ▾	REMOTE, HYBRID, ONSITE.
employment_type	VarChar(20) ▾	FULL_TIME, PART_TIME, etc..
salary	VarChar(100) ▾	Compensation estimation text.
promoted	Boolean ▾	Sponsored listing flag.
created_at	DateTime ▾	Creation timestamp.

3. Applications Table

Tracks submissions made by candidates to specific job listings.

Field	Type & Constraints	Description
id	BigInt, PK, Auto-inc	Unique application ID.
job_id	BigInt, FK (jobs)	Application target listing.
candidate_id	BigInt, FK (users)	Applicant user record.
status	VarChar(20)	PENDING, ACCEPTED, REJECTED.
applied_at	DateTime	Submission timestamp.

4. Experience & Education Tables

These tables capture the professional and academic background of the users.

Table	Primary Fields	Key Details
experiences	company, role, dates	Professional history and highlights.
educations	school, degree, field	Academic records and achievements.
certifications	name, issuer, date	Verified licenses and certificates.
skills	name, endorsement_count	Key professional keywords.

5. Recommendations & Saved Jobs

Captures peer feedback and user engagement within the platform.

- **Recommendations:** Stores testimonials, including the author's name, title, and the recommendation content.
- **User Saved Jobs (Join Table):** A many-to-many link between `users` and `jobs` for bookmarked listings.